Chapter Review

Key Terms

absolute value function $f(x) = \begin{cases} -x, x < 0 \\ x, x \geq 0 \end{cases}$

algebraic function a function involving any combination of only the basic operations of addition, subtraction, multiplication, division, powers, and roots applied to an input variable x

base the number b in the exponential function $f(x) = b^x$ and the logarithmic function $f(x) = \log_b x$

composite function given two functions f and g, a new function, denoted $g \circ f$, such that $(g \circ f)(x) = g(f(x))$

cubic function a polynomial of degree 3; that is, a function of the form $f(x) = ax^3 + bx^2 + cx + d$, where $a \neq 0$

decreasing on the interval I a function decreasing on the interval I if, for all $x_1, x_2 \in I$, $f(x_1) \geq f(x_2)$ if $x_1 < x_2$

degree for a polynomial function, the value of the largest exponent of any term

dependent variable the output variable for a function

domain the set of inputs for a function

even function a function is even if $f(-x) = f(x)$ for all x in the domain of f

exponent the value x in the expression b^x

function a set of inputs, a set of outputs, and a rule for mapping each input to exactly one output

graph of a function the set of points (x, y) such that x is in the domain of f and $y = f(x)$

horizontal line test a function f is one-to-one if and only if every horizontal line intersects the graph of f, at most, once

hyperbolic functions the functions denoted $\sinh, \cosh, \tanh, \text{csch}, \text{sech}$, and $\coth$, which involve certain combinations of e^x and e^{-x}

increasing on the interval I a function increasing on the interval I if for all $x_1, x_2 \in I$, $f(x_1) \leq f(x_2)$ if $x_1 < x_2$

independent variable the input variable for a function

inverse function for a function f, the inverse function f^{-1} satisfies $f^{-1}(y) = x$ if $f(x) = y$

inverse hyperbolic functions the inverses of the hyperbolic functions where $\cosh$ and sech are restricted to the domain $\left[0, \infty\right)$; each of these functions can be expressed in terms of a composition of the natural logarithm function and an algebraic function

inverse trigonometric functions the inverses of the trigonometric functions are defined on restricted domains where they are one-to-one functions

linear function a function that can be written in the form $f(x) = mx + b$

logarithmic function a function of the form $f(x) = \log_b(x)$ for some base $b > 0, b \neq 1$ such that $y = \log_b(x)$ if and only if $b^y = x$

mathematical model A method of simulating real-life situations with mathematical equations

natural exponential function the function $f(x) = e^x$

natural logarithm the function $\ln x = \log_e x$

number e as m gets larger, the quantity $(1 + (1/m))^m$ gets closer to some real number; we define that real number to be e; the value of e is approximately 2.718282

odd function a function is odd if $f(-x) = -f(x)$ for all x in the domain of f

one-to-one function a function f is one-to-one if $f(x_1) \neq f(x_2)$ if $x_1 \neq x_2$

periodic function a function is periodic if it has a repeating pattern as the values of x move from left to right

piecewise-defined function a function that is defined differently on different parts of its domain

point-slope equation equation of a linear function indicating its slope and a point on the graph of the function

polynomial function a function of the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$

power function a function of the form $f(x) = x^n$ for any positive integer $n \geq 1$

quadratic function a polynomial of degree 2; that is, a function of the form $f(x) = ax^2 + bx + c$ where $a \neq 0$

radians for a circular arc of length s on a circle of radius 1, the radian measure of the associated angle θ is s

range the set of outputs for a function

rational function a function of the form $f(x) = p(x)/q(x)$, where $p(x)$ and $q(x)$ are polynomials

restricted domain a subset of the domain of a function f

root function a function of the form $f(x) = x^{1/n}$ for any integer $n \geq 2$

slope the change in y for each unit change in x

slope-intercept form equation of a linear function indicating its slope and y-intercept

symmetry about the origin the graph of a function f is symmetric about the origin if $(-x, -y)$ is on the graph of f whenever (x, y) is on the graph

symmetry about the y-axis the graph of a function f is symmetric about the y-axis if $(-x, y)$ is on the graph of f whenever (x, y) is on the graph

table of values a table containing a list of inputs and their corresponding outputs
transcendental function a function that cannot be expressed by a combination of basic arithmetic operations
transformation of a function a shift, scaling, or reflection of a function
trigonometric functions functions of an angle defined as ratios of the lengths of the sides of a right triangle
trigonometric identity an equation involving trigonometric functions that is true for all angles θ for which the functions in the equation are defined
vertical line test given the graph of a function, every vertical line intersects the graph, at most, once
zeros of a function when a real number x is a zero of a function f, $f(x) = 0$

Key Equations

Composition of two functions	$(g \circ f)(x) = g(f(x))$
Absolute value function	$f(x) = \begin{cases} -x, x < 0 \\ x, x \geq 0 \end{cases}$
Point-slope equation of a line	$y - y_1 = m(x - x_1)$
Slope-intercept form of a line	$y = mx + b$
Standard form of a line	$ax + by = c$
Polynomial function	$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$
Generalized sine function	$f(x) = A \sin(B(x - \alpha)) + C$
Inverse functions	$f^{-1}(f(x)) = x$ for all x in D, and $f\left(f^{-1}(y)\right) = y$ for all y in R.

Key Concepts

1.1 Review of Functions

- A function is a mapping from a set of inputs to a set of outputs with exactly one output for each input.
- If no domain is stated for a function $y = f(x)$, the domain is considered to be the set of all real numbers x for which the function is defined.
- When sketching the graph of a function f, each vertical line may intersect the graph, at most, once.
- A function may have any number of zeros, but it has, at most, one y-intercept.
- To define the composition $g \circ f$, the range of f must be contained in the domain of g.
- Even functions are symmetric about the y-axis whereas odd functions are symmetric about the origin.

1.2 Basic Classes of Functions

- The power function $f(x) = x^n$ is an even function if n is even and $n \neq 0$, and it is an odd function if n is odd.
- The root function $f(x) = x^{1/n}$ has the domain $[0, \infty)$ if n is even and the domain $(-\infty, \infty)$ if n is odd. If n is odd, then $f(x) = x^{1/n}$ is an odd function.
- The domain of the rational function $f(x) = p(x)/q(x)$, where $p(x)$ and $q(x)$ are polynomial functions, is the set of x such that $q(x) \neq 0$.
- Functions that involve the basic operations of addition, subtraction, multiplication, division, and powers are algebraic functions. All other functions are transcendental. Trigonometric, exponential, and logarithmic functions are examples of transcendental functions.
- A polynomial function f with degree $n \geq 1$ satisfies $f(x) \to \pm\infty$ as $x \to \pm\infty$. The sign of the output as $x \to \infty$ depends on the sign of the leading coefficient only and on whether n is even or odd.
- Vertical and horizontal shifts, vertical and horizontal scalings, and reflections about the x- and y-axes are examples of transformations of functions.

1.3 Trigonometric Functions

- Radian measure is defined such that the angle associated with the arc of length 1 on the unit circle has radian

measure 1. An angle with a degree measure of $180°$ has a radian measure of π rad.

- For acute angles θ, the values of the trigonometric functions are defined as ratios of two sides of a right triangle in which one of the acute angles is θ.
- For a general angle θ, let (x, y) be a point on a circle of radius r corresponding to this angle θ. The trigonometric functions can be written as ratios involving x, y, and r.
- The trigonometric functions are periodic. The sine, cosine, secant, and cosecant functions have period 2π. The tangent and cotangent functions have period π.

1.4 Inverse Functions

- For a function to have an inverse, the function must be one-to-one. Given the graph of a function, we can determine whether the function is one-to-one by using the horizontal line test.
- If a function is not one-to-one, we can restrict the domain to a smaller domain where the function is one-to-one and then define the inverse of the function on the smaller domain.
- For a function f and its inverse f^{-1}, $f\left(f^{-1}(x)\right) = x$ for all x in the domain of f^{-1} and $f^{-1}(f(x)) = x$ for all x in the domain of f.
- Since the trigonometric functions are periodic, we need to restrict their domains to define the inverse trigonometric functions.
- The graph of a function f and its inverse f^{-1} are symmetric about the line $y = x$.

1.5 Exponential and Logarithmic Functions

- The exponential function $y = b^x$ is increasing if $b > 1$ and decreasing if $0 < b < 1$. Its domain is $(-\infty, \infty)$ and its range is $\left(0, \infty\right)$.
- The logarithmic function $y = \log_b(x)$ is the inverse of $y = b^x$. Its domain is $\left(0, \infty\right)$ and its range is $\left(-\infty, \infty\right)$.
- The natural exponential function is $y = e^x$ and the natural logarithmic function is $y = \ln x = \log_e x$.
- Given an exponential function or logarithmic function in base a, we can make a change of base to convert this function to any base $b > 0, b \neq 1$. We typically convert to base e.
- The hyperbolic functions involve combinations of the exponential functions e^x and e^{-x}. As a result, the inverse hyperbolic functions involve the natural logarithm.

Review Exercises

True or False? Justify your answer with a proof or a counterexample.

310. A function is always one-to-one.

311. $f \circ g = g \circ f$, assuming f and g are functions.

312. A relation that passes the horizontal and vertical line tests is a one-to-one function.

313. A relation passing the horizontal line test is a function.

For the following problems, state the domain and range of the given functions:

$f = x^2 + 2x - 3, \qquad g = \ln(x - 5), \qquad h = \frac{1}{x+4}$

314. h

315. g

316. $h \circ f$

317. $g \circ f$

Find the degree, y-intercept, and zeros for the following polynomial functions.

318. $f(x) = 2x^2 + 9x - 5$

319. $f(x) = x^3 + 2x^2 - 2x$

Simplify the following trigonometric expressions.

320. $\frac{\tan^2 x}{\sec^2 x} + \cos^2 x$

321. $\cos^2 x - \sin^2 x$

Solve the following trigonometric equations on the interval $x = [-2\pi, 2\pi]$ exactly.

322. $6\cos^2 x - 3 = 0$

323. $\sec^2 x - 2\sec x + 1 = 0$

Solve the following logarithmic equations.

324. $5^x = 16$

325. $\log_2(x + 4) = 3$

Are the following functions one-to-one over their domain of existence? Does the function have an inverse? If so, find the inverse $f^{-1}(x)$ of the function. Justify your answer.

326. $f(x) = x^2 + 2x + 1$

327. $f(x) = \frac{1}{x}$

For the following problems, determine the largest domain on which the function is one-to-one and find the inverse on that domain.

328. $f(x) = \sqrt{9 - x}$

329. $f(x) = x^2 + 3x + 4$

330. A car is racing along a circular track with diameter of 1 mi. A trainer standing in the center of the circle marks his progress every 5 sec. After 5 sec, the trainer has to turn 55° to keep up with the car. How fast is the car traveling?

For the following problems, consider a restaurant owner who wants to sell T-shirts advertising his brand. He recalls that there is a fixed cost and variable cost, although he does not remember the values. He does know that the T-shirt printing company charges $440 for 20 shirts and $1000 for 100 shirts.

331. a. Find the equation $C = f(x)$ that describes the total cost as a function of number of shirts and b. determine how many shirts he must sell to break even if he sells the shirts for $10 each.

332. a. Find the inverse function $x = f^{-1}(C)$ and describe the meaning of this function. b. Determine how many shirts the owner can buy if he has $8000 to spend.

For the following problems, consider the population of Ocean City, New Jersey, which is cyclical by season.

333. The population can be modeled by $P(t) = 82.5 - 67.5\cos[(\pi/6)t]$, where t is time in months ($t = 0$ represents January 1) and P is population (in thousands). During a year, in what intervals is the population less than 20,000? During what intervals is the population more than 140,000?

334. In reality, the overall population is most likely increasing or decreasing throughout each year. Let's reformulate the model as $P(t) = 82.5 - 67.5\cos[(\pi/6)t] + t$, where t is time in months ($t = 0$ represents January 1) and P is population (in thousands). When is the first time the population reaches 200,000?

For the following problems, consider radioactive dating. A human skeleton is found in an archeological dig. Carbon dating is implemented to determine how old the skeleton is by using the equation $y = e^{rt}$, where y is the ratio of radiocarbon still present in the material, t is the number of years passed, and $r = -0.0001210$ is the decay rate of radiocarbon.

335. If the skeleton is expected to be 2000 years old, what percentage of radiocarbon should be present?

336. Find the inverse of the carbon-dating equation. What does it mean? If there is 25% radiocarbon, how old is the skeleton?